

Is Florida getting warmer?

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We have a time record of the average temperature in Florida over the years. This time series is plotted on Figure 1 where one can observe there seems to be a trend of rising temperature in Florida between. However, we want to make sure that the positive correlation between time and temperature is significant. We will therefore perform a simple statistical test, using correlation of randomized value of year and temperature as *null hypothesis*. If a significant amount of random correlations are above the effective correlation between time and temperature, then it means that the difference with the null hypothesis is not significant. If the null hypothesis correlation set is large enough, the ratio of random correlations that are above the effective one is referred to as *p-value*.

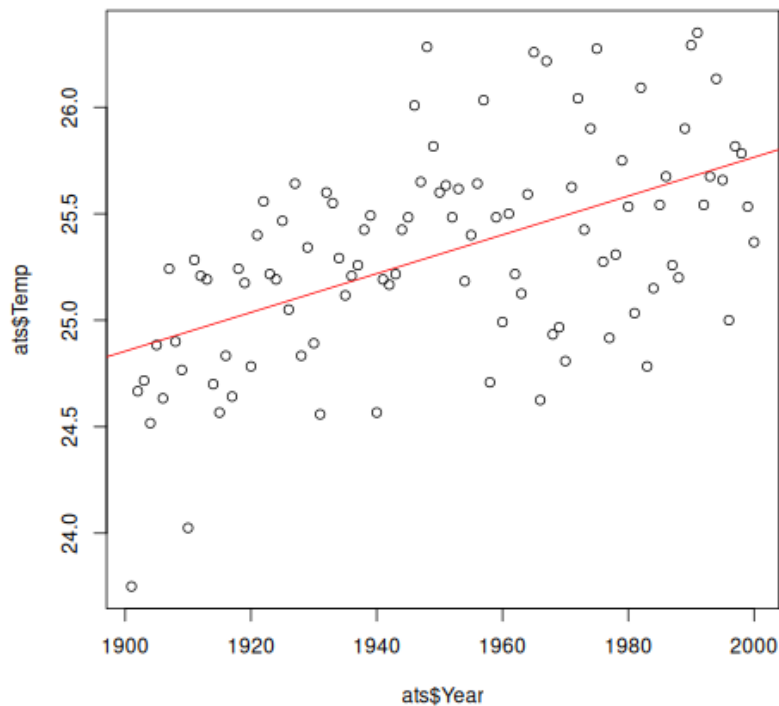


Figure 1: Time series of Florida temperature with linear regression line in red.

We therefore find out that there is a significant (p value is close to 0) positive correlation (0.533) between time and temperature, suggesting that Florida is indeed getting warmer.